

Time Series Exercise Sheet 13

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Exercise 13.1

Consider two LTI filters L_1 and L_2 , and let $\alpha \in \mathbb{C}$ then the filter

$$L = \alpha L_1 + L_2 \tag{1}$$

is also an LTI filter.

Solution 13.1

Let $\{X_t\}$, $\{Y_t\}$ be sequences, and $\beta \in \mathbb{C}$, then

$$\begin{aligned} L[\beta\{X_t\} + \{Y_t\}] &= \alpha L_1[\beta\{X_t\} + \{Y_t\}] + L_2[\beta\{X_t\} + \{Y_t\}] \\ &= \alpha\beta L_1[\{X_t\}] + \alpha L_1[\{Y_t\}] + \beta L_2[\{X_t\}] + L_2[\{Y_t\}] \quad (\text{linearity of LTI filters}) \\ &= \beta L[\{X_t\}] + L[\{Y_t\}]. \end{aligned}$$

Now for time invariance,

$$\begin{aligned} L[B[\{X_t\}]] &= \alpha L_1[B[\{X_t\}]] + L_2[B[\{X_t\}]] \\ &= B[\alpha L_1[\{X_t\}]] + B[L_2[\{X_t\}]] \quad (\text{linearity of LTI filters}) \\ &= B[L[\{X_t\}]]. \end{aligned}$$

Exercise 13.2

Consider two LTI filters L_1 and L_2 . The filter $L = L_1 L_2$, i.e. so that

$$L[\{X_t\}] = L_1[L_2[\{X_t\}]] \tag{2}$$

is also an LTI filter.

Solution 13.2

Let $\{X_t\}$, $\{Y_t\}$ be sequences, and $\alpha \in \mathbb{C}$, then

$$\begin{aligned} L[\alpha\{X_t\} + \{Y_t\}] &= L_1[L_2[\alpha\{X_t\} + \{Y_t\}]] \\ &= L_1[\alpha L_2[\{X_t\}] + L_2[\{Y_t\}]] \\ &= \alpha L_1[L_2[\{X_t\}]] + L_1[L_2[\{Y_t\}]] \\ &= \alpha L[\{X_t\}] + L[\{Y_t\}]. \end{aligned}$$

Now for time invariance

$$\begin{aligned} L[B[\{X_t\}]] &= L_1[L_2[B[\{X_t\}]]] \\ &= L_1[B[L_2[\{X_t\}]]] \\ &= B[L_1[L_2[\{X_t\}]]]. \end{aligned}$$

Exercise 13.3

Consider deterministic input sequences in ℓ^1 . Let $L : \ell^1 \rightarrow \mathbb{C}^{\mathbb{Z}}$ be a digital filter such that, for each fixed $u \in \mathbb{Z}$, the map

$$\{X_t\} \mapsto L[\{X_t\}]_u$$

is continuous with respect to the ℓ^1 norm. Define the impulse response by

$$h_m = L[\{\delta_{t,-m}\}]_0,$$

and assume $h \in \ell^1$.

Prove that L is LTI on ℓ^1 (that is, linear and time-invariant for all inputs in ℓ^1) if and only if, for every $\{X_t\} \in \ell^1$,

$$L[\{X_t\}]_u = \sum_{m \in \mathbb{Z}} h_{u-m} X_m \quad (3)$$

for any $u \in \mathbb{Z}$. In your proof, justify carefully all limit/sum interchanges.

Solution 13.3

($\Rightarrow$) Assume that L is LTI on ℓ^1 . For $N \in \mathbb{N}$, define the truncated sequence

$$X_t^{(N)} = \sum_{|m| \leq N} X_m \delta_{t,m}.$$

Since $\{X_t\} \in \ell^1$, we have $\|X^{(N)} - X\|_1 \rightarrow 0$. Therefore, by continuity of $\{X_t\} \mapsto L[\{X_t\}]_u$,

$$L[\{X_t\}]_u = \lim_{N \rightarrow \infty} L[\{X_t^{(N)}\}]_u.$$

For each N , linearity gives a finite sum:

$$L[\{X_t^{(N)}\}]_u = \sum_{|m| \leq N} X_m L[\{\delta_{t,m}\}]_u.$$

By time invariance,

$$L[\{\delta_{t,m}\}]_u = L[\{\delta_{t,m-u}\}]_0 = h_{u-m},$$

so

$$L[\{X_t^{(N)}\}]_u = \sum_{|m| \leq N} h_{u-m} X_m.$$

Also, since $h \in \ell^1$, we have $\|h\|_\infty \leq \|h\|_1$, hence

$$\sum_{m \in \mathbb{Z}} |h_{u-m} X_m| \leq \|h\|_\infty \sum_{m \in \mathbb{Z}} |X_m| < \infty.$$

Therefore the partial sums converge to the full series, and

$$L[\{X_t\}]_u = \lim_{N \rightarrow \infty} \sum_{|m| \leq N} h_{u-m} X_m = \sum_{m \in \mathbb{Z}} h_{u-m} X_m.$$

($\Leftarrow$) Assume there exists $h \in \ell^1$ such that, for every $\{X_t\} \in \ell^1$,

$$L[\{X_t\}]_u = \sum_{m \in \mathbb{Z}} h_{u-m} X_m.$$

The series is absolutely convergent because

$$\sum_{m \in \mathbb{Z}} |h_{u-m} X_m| \leq \|h\|_{\infty} \|X\|_1 \leq \|h\|_1 \|X\|_1.$$

For $\alpha \in \mathbb{C}$, and any $u \in \mathbb{Z}$,

$$\begin{aligned} L[\alpha \{X_t\} + \{Y_t\}]_u &= \sum_{m \in \mathbb{Z}} h_{u-m} (\alpha X_m + Y_m) \\ &= \alpha \sum_{m \in \mathbb{Z}} h_{u-m} X_m + \sum_{m \in \mathbb{Z}} h_{u-m} Y_m \\ &= \alpha L[\{X_t\}]_u + L[\{Y_t\}]_u. \end{aligned}$$

So L is linear.

For time invariance, using absolute convergence to justify reindexing,

$$\begin{aligned} L[B[\{X_t\}]]_u &= \sum_{m \in \mathbb{Z}} h_{u-m} X_{m-1} \\ &= \sum_{m' \in \mathbb{Z}} h_{u-1-m'} X_{m'} \quad (m' = m - 1) \\ &= L[\{X_t\}]_{u-1} \\ &= B[L[\{X_t\}]]_u. \end{aligned}$$

Hence L is time-invariant on ℓ^1 , and therefore LTI on ℓ^1 .

Exercise 13.4

Consider a stationary mean-zero time series $\{X_t\}$, with spectral representation

$$X_t = \int_{-1/2}^{1/2} e^{2\pi i f t} dZ(f). \quad (4)$$

Assume that we observe this time series at the points $1, \dots, n$, and we define the tapered discrete Fourier transform by

$$J_h(f) = \sum_{t=1}^n h_t X_t e^{-2\pi i f t} \quad (5)$$

where $\|h_t\|_2^2 = 1$ (we assume the mean is known to be zero, so do no mean correction). Show that

$$J_h(f) = \int_{-1/2}^{1/2} H(f - f') dZ(f'). \quad (6)$$

Solution 13.4

We see from the definition that

$$\begin{aligned} J_h(f) &= \sum_{t=1}^n h_t X_t e^{-2\pi i f t} \\ &= \sum_{t=1}^n h_t \int_{-1/2}^{1/2} e^{2\pi i f' t} dZ(f') e^{-2\pi i f t} \\ &= \int_{-1/2}^{1/2} \sum_{t=1}^n h_t e^{2\pi i (f' - f) t} dZ(f') \\ &= \int_{-1/2}^{1/2} H(f - f') dZ(f'). \end{aligned}$$

Exercise 13.5

If $\{X_t\}$ is a stationary series, define $\{Y_t\} = (I - B)[\{X_t\}]$. Is Y_t stationary? If so, what is the spectral density function of $\{Y_t\}$ in terms of the spectral density function of $\{X_t\}$?

Solution 13.5

Note that $(I - B)$ is an LTI filter with impulse response in ℓ_1 , so from Theorem 9.11 $\{Y_t\}$ is stationary. Now we may use Theorem 9.12 to compute the spectral density function. In particular, the transfer function of this filter is given by

$$H(f) = 1 - e^{-2\pi if}, \quad (7)$$

and so

$$\begin{aligned} S_Y(f) &= |1 - e^{-2\pi if}|^2 S_X(f) \\ &= 2(1 - \cos(2\pi f)) S_X(f) \\ &= 4 \sin^2(\pi f) S_X(f). \end{aligned}$$

Exercise 13.6

In lecture 3, we claimed that AR(p) processes were time reversible. In other words, If $\{X_t\}$ is an AR(p) process, then if $\{Y_t\}$ is such that for all $t \in \mathbb{Z}$, $Y_t = X_{-t}$, then Y_t is an AR(p) process with the same parameters as X_t . Specifically, if X_t had an AR representation

$$X_t = \sum_{j=1}^p \phi_j X_{t-j} + \epsilon_t \quad (8)$$

then

$$Y_t = \sum_{j=1}^p \phi_j Y_{t-j} + \tilde{\nu}_t \quad (9)$$

where $\tilde{\nu}_t$ has the same distribution as ϵ_t . Prove this result.

Solution 13.6

Let $\phi_0 = -1$, and define

$$\tilde{\nu}_t = \sum_{j=0}^p \phi_j Y_{t-j}.$$

All that remains is to show that $\tilde{\nu}_t$ is Gaussian white noise with mean zero and variance σ_ϵ^2 . Clearly it is Gaussian as it is a linear combination of Gaussians. Furthermore, $\mathbb{E}[X_t] = 0$, so $\mathbb{E}[\tilde{\nu}_t] = 0$ by linearity.

Now, we see that for all $\tau \in \mathbb{Z}$

$$\begin{aligned} \gamma_\tau^{(Y)} &= \mathbb{E}[Y_t Y_{t+\tau}] \\ &= \mathbb{E}[X_{-t} X_{-t-\tau}] \\ &= \gamma_{-\tau}^{(X)} \\ &= \gamma_\tau^{(X)}. \end{aligned}$$

Therefore $S_Y(f) = S_X(f)$ for all $f \in \mathbb{R}$. Finally, we see that $\{\tilde{\nu}_t\}$ results from applying an LTI filter to $\{Y_t\}$. In particular, from Lemma 9.13 and Theorem 9.12, we have

$$\begin{aligned} S_{\tilde{\nu}}(f) &= |\Phi(e^{-2\pi if})|^2 S_Y(f) \\ &= |\Phi(e^{-2\pi if})|^2 S_X(f) \\ &= \sigma_\epsilon^2 \left| \frac{\Phi(e^{-2\pi if})}{\Phi(e^{-2\pi if})} \right|^2 \\ &= \sigma_\epsilon^2 \end{aligned}$$

where the penultimate line follows from Theorem 9.14. Therefore, we know that $\tilde{\nu}$ is white noise with variance σ_ϵ^2 .